

# Problem Statement No. 5

## Travel Planner Agent

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### The Challenge

A Travel Planner Agent is an AI-powered assistant that helps users plan trips efficiently and intelligently. It uses real-time data to suggest destinations, build itineraries, and recommend transport and accommodation options. By understanding user preferences, budgets, and constraints, it tailors personalized travel plans. Integrated with maps, weather updates, and local guides, it ensures a smooth travel experience. The agent can also manage bookings, alert users to changes, and optimize schedules on the go. This smart assistant transforms complex travel planning into a seamless, enjoyable process.

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### Technology

Use of IBM Cloud Lite services / IBM Granite is mandatory.

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### Our Solution - AI Travel Planner

Built with IBM Watsonx.ai (Meta Llama 3.3 70B / Mistral) and Python Flask. 10 specialised AI agents run in parallel via ThreadPoolExecutor: Destination Recommender, Itinerary Planner, Budget Planner, Transport Advisor, Hotel Recommender, Weather Advisory, Local Guide, Restaurant Advisor, Packing Checklist, and Chat Assistant. Fully personalised based on user preferences, budget, and travel style. Includes PDF export, in-memory trip store, and off-topic chat guard. Accessible via browser at localhost:5000.